Name	Subject Class	Class	Index Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2021

BIOLOGY **9744/02**

26 August 2021

HIGHER 2 **2 hours**

Paper 2 Structured Questions
READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on the top of this page.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
No additional materials are required.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
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8	
9	
10	
11	
Total	**100**

This question paper consists of **28** printed pages.

1 Human pancreatic cells produce proteases that act on the contents of the small intestine. An experiment was performed to track the pathway of protease synthesis and release. This involved incubating pancreatic cells for a brief period with amino acids that had been radioactively labelled. Radioactively-labelled proteins were detected in various organelles 120 minutes after incubation.

(a)(i) State two organelles which are likely to be found in abundance in a human pancreatic cell and outline the specific function of each named organelle in the cell.

[4]

(ii) A student predicts that radioactively-labelled proteases will be detected at the nucleus 120 minutes after incubation because it is involved in protein synthesis.

Explain why the student's prediction is incorrect.

[2]

Fig. 1.1 shows the cross-section of a cell surface membrane.

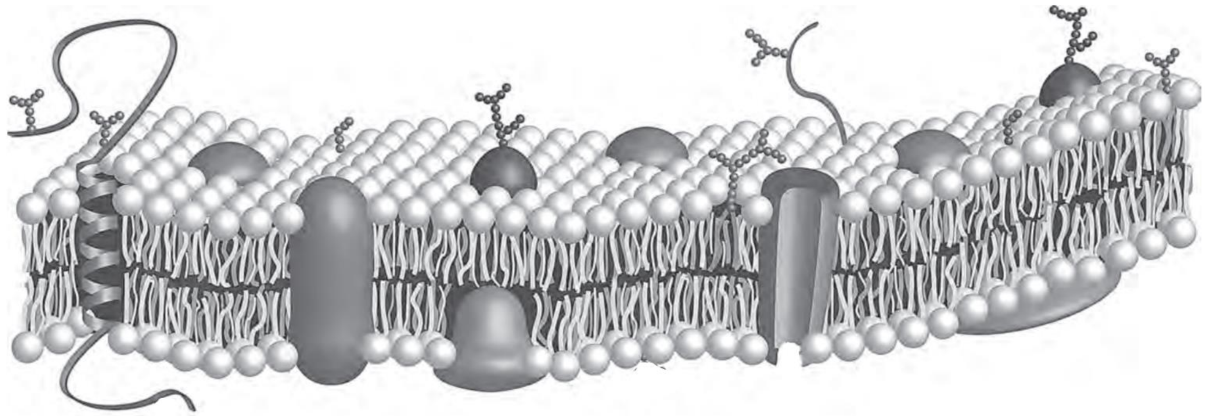

Fig. 1.1

(b)(i) There is a higher Na^+ concentration in the extracellular environment than in the intracellular environment.

Draw an arrow on Fig. 1.1 to show how Na^+ diffuse across the cell surface membrane. [1]

(ii) Explain your answer in **(b)(i)**.

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.... [2]

[Total: 9]

2 Progressive telomere shortening has been linked to aging. In a study, telomere length in red blood cells of zebra finches is measured and recorded throughout their natural lifespan. The first measurement (shown as year = 0) was taken at 25 days, and measurements were taken at various points thereafter as shown in Fig. 2.1. Mean telomere length is represented as a ratio relative to a standard reference point.

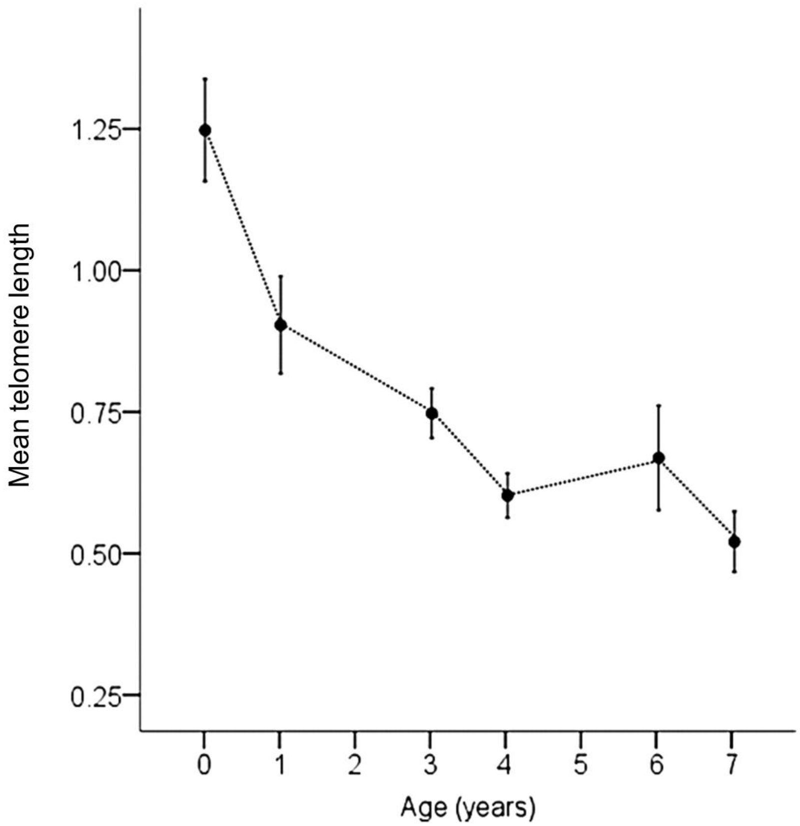

Fig. 2.1

(a)(i) Explain the relationship between telomere length and age of zebra finches.

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(ii) Compare the structures of telomeres and centromeres.

[2]

In embryonic stem (ES) cells, telomere length is maintained and the cells can continue to divide for long periods of time. This property is known as self-renewal.

(b)(i) Explain the significance of the ability of stem cells to renew themselves in living organisms.

[2]

(ii) Suggest why telomeres are not shortened in ES cells despite numerous rounds of cell division.

[1]

[Total: 8]

For Examiner's Use

3 Fig. 3.1 is an illustration of the steps in DNA replication in prokaryotes.

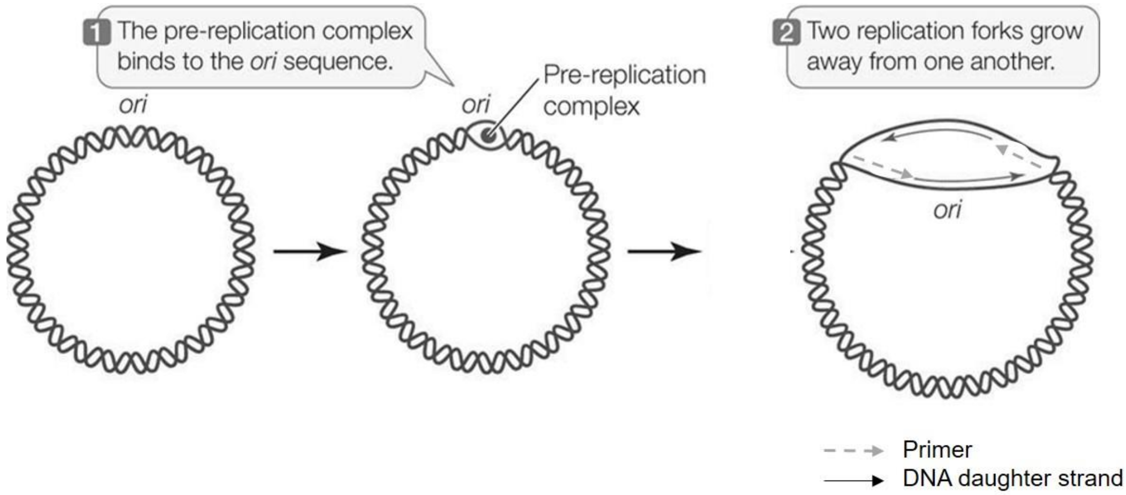

Fig. 3.1

(a) Define the term "ori" in Fig. 3.1.

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.... [1]

(b) Primers and DNA daughter strands consist of different types of nucleotides.

Describe one way in which these nucleotides are similar in structure and explain why this feature is important for their functions.

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(c) With reference to Fig. 3.1, state and explain an error in this illustration of DNA replication in prokaryotes.

[3]

Mutations in the *top* gene (the structural gene coding for *E.coli* DNA topoisomerase) may result in a number of impacts on DNA replication. In addition, it also has impacts on transcription.

(d) Suggest why loss-of-function mutations in the *top* gene can also impact transcription.

[3]

[Total: 9]

4 The fragile X mental retardation protein (FMRP) is essential in regulating the synthesis of neurone proteins which are important for normal brain function. Patients with fragile X syndrome (FXS) have very high number of CGG repeats at the beginning of the *FMRP* gene. These CGG repeats cause the DNA to fold into hairpins.

Fig 4.1 shows the expression of *FMRP* gene in a normal individual and in a patient with FXS.

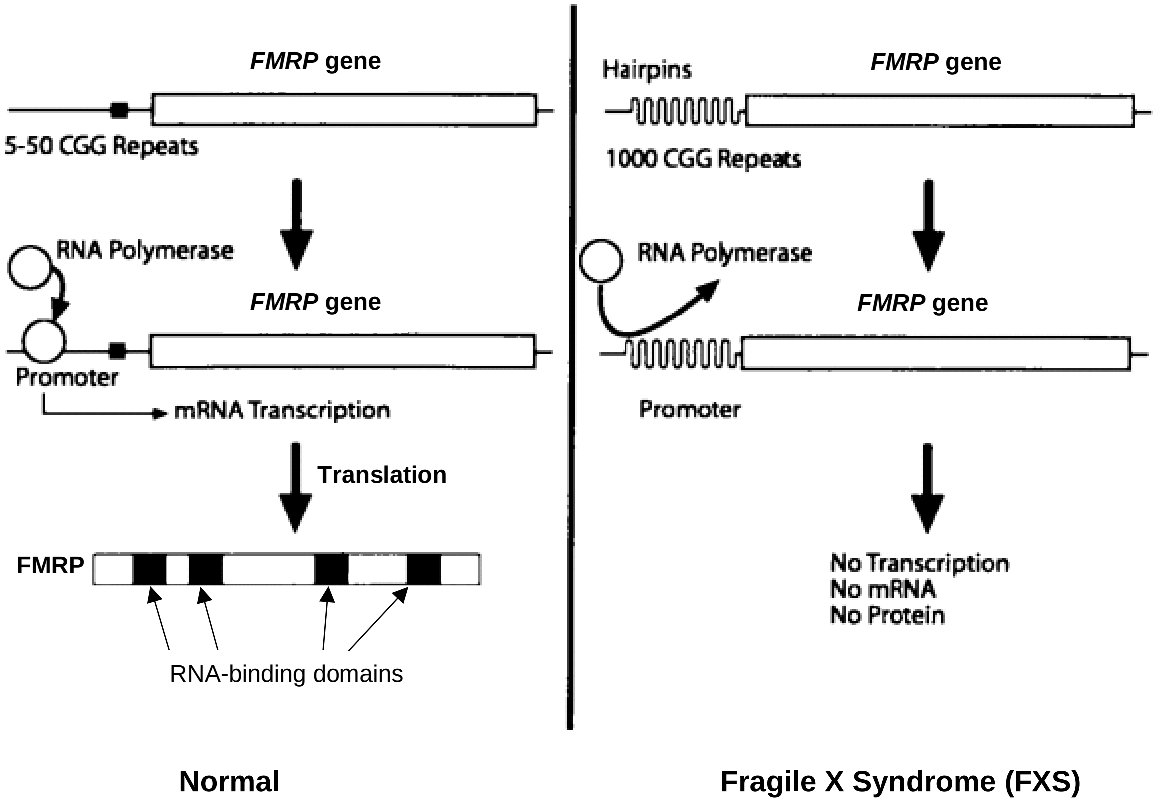

Fig. 4.1

(a) Explain the significance of the hairpin loops caused by the CGG repeats.

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FMRP has RNA-binding domains which binds to mRNA coding for neurone proteins, preventing protein synthesis. The absence of functional FMRP leads to an increased rate of synthesis of neurone proteins which in turn results in FXS.

(b) State which level of control of gene expression FMRP is involved in.

[1]

(c) Some patients have a mild version of FXS, with 200-300 CGG repeats at the start of the *FMRP* gene. In these patients, a low number of FMRP can still be produced.

A student suggested that one possible way of treating mild FXS is to introduce enzymes which lengthen the poly-A tail of the *FMRP* mRNA. Explain if the student is right.

[3]

(d) Patients with FXS are able to produce normal gametes as meiosis is not affected by this condition.

Outline how haploid gametes are formed during meiosis.

[2]

[Total: 9]

5 A study was done to measure the expression of the structural genes of the *lac* operon in various mutant bacterial strain in the absence of lactose. The bacterial strains *lacI*$^{+}$ and *lacI*$^{-}$ have the regulatory gene *lacI* present and deleted respectively. A positive measurement refers to the presence of the products in Table 5.1.

Table 5.1

strain	amount of β-galactosidase/ AU	amount of lactose permease/ AU
lacI^{+}	0.005	0
lacI^{-}	2.022	0.1238
lacI^{-} *lacY*$^{-}$	1.082	0

(a) With reference to the *lac* operon, distinguish between a structural gene and a regulatory gene.

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(b) Explain the difference in the amount of β-galactosidase in the bacterial strains *lacI*$^{+}$ *and lacI*$^{-}$.

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(c) In the bacterial strain *lacI*⁺, minute amount of β-galactosidase could still be detected, unlike lactose permease which was undetected. Suggest why the amount of lactose permease is not measurable in some of the bacteria strains.

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.... [1]

In the presence of glucose, bacteria strains preferably use glucose as the respiratory substrate. Glucose, stored in the form of glycogen in the liver, is also an important respiratory substrate in humans.

(d) Describe how the structure of glycogen relates to its function as an energy storage molecule.

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[Total: 10]

For Examiner's Use

6 Bacteria is a diverse group of organisms. Bacteria species can be categorised into Gram-negative or Gram-positive. Fig. 6.1 shows the membranes and cell walls of these two types of bacteria.

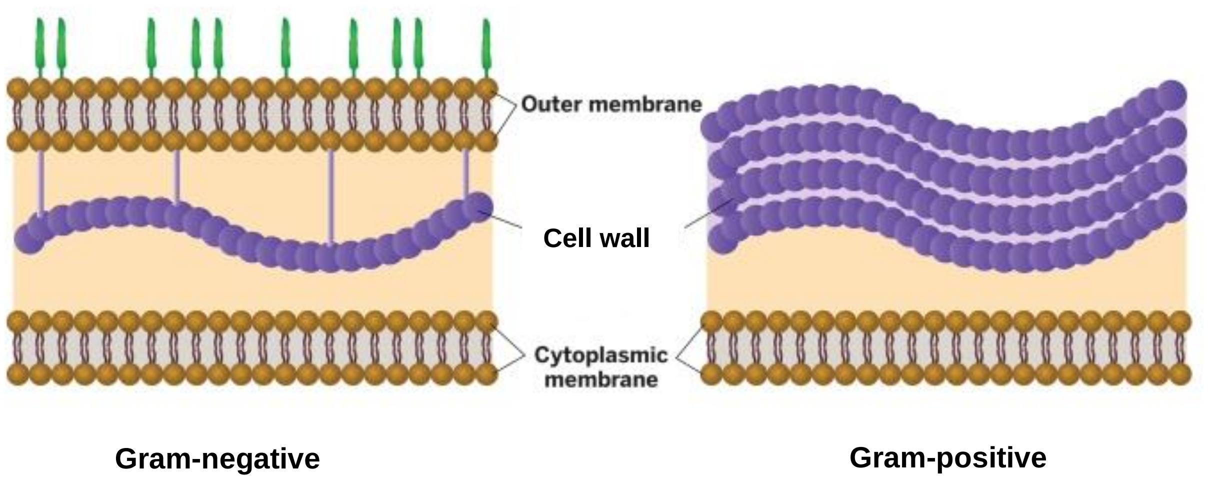

Fig. 6.1

(a) Name the main component of the bacterial cell wall.

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Unlike eukaryotic organisms, bacteria do not contain mitochondria but are still able to carry out respiration. Fig. 6.2 shows the different proteins embedded within the membranes in a Gram-negative bacteria.

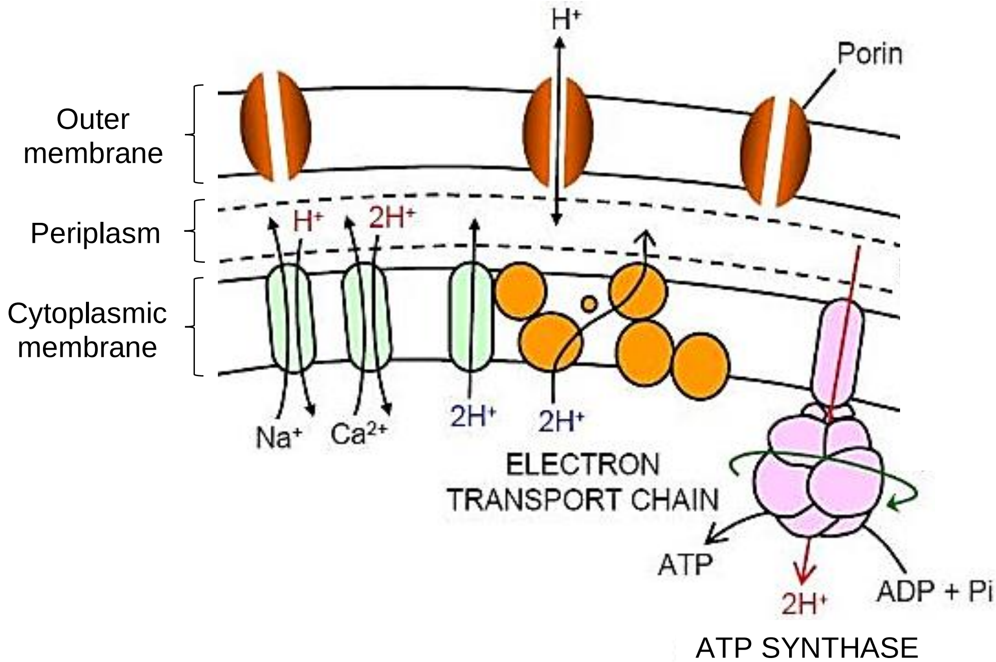

Fig. 6.2

(b)(i) With reference to Fig. 6.2, explain how the structure of the membrane allows the bacteria to undergo respiration.

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For Examiner's Use

(ii) The Endosymbiotic Theory states that eukaryotic organelles like the mitochondria originated from prokaryotic microbes. The prokaryotic microbe was thought to be engulfed by a eukaryotic cell.

It was hypothesised that the prokaryotic microbe related to the Endosymbiotic Theory was a Gram-negative bacteria. Discuss if the information above supports this.

[2]

(c) Some bacteria can respire anaerobically in low oxygen conditions. Bacteria respiring in low oxygen conditions produce lactate.

Outline the processes leading to the formation of lactate.

[2]

[Total: 9]

7 *Escherichia coli* is a common human food contaminant. An *E. coli* strain carrying the *aphA1* and *aphA2* genes is resistant to the antibiotic kanamycin.

(a) Describe the conditions needed for *aphA1* and *aphA2* genes to be transferred to other *E. coli* cells by transformation.

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.... [3]

(b) The kanamycin-resistant *E. coli* strain can transfer the *aphA1* and *aphA2* genes to non-kanamycin-resistant *E. coli* via other mechanisms, such as transduction and conjugation.

Contrast conjugation and transduction.

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.... [3]

Horizontal gene transfer between bacteria can cause undesirable spread of diseases and antibiotic resistance. Multidrug-resistance tuberculosis (MDR-TB) is caused by a strain of *M. tuberculosis* that is resistant to the two antibiotics, isoniazid and rifampicin.

Fig. 7.1 shows the number of cases of MDR-TB and the number of antibiotic prescriptions administered to individuals between 2000 and 2010.

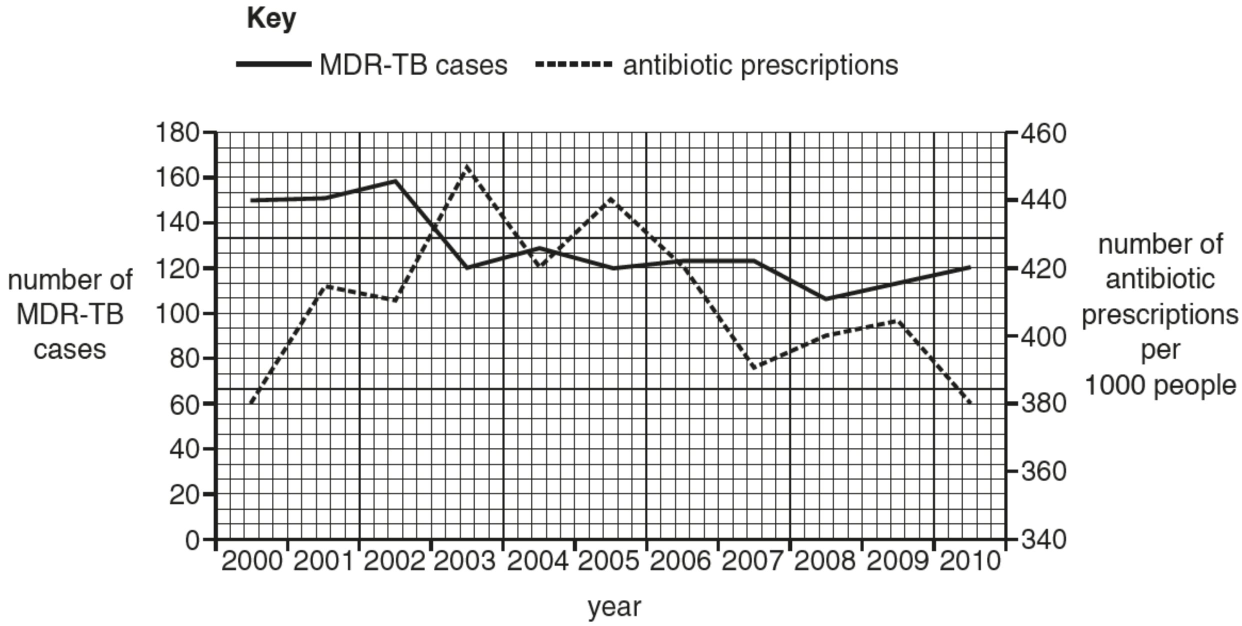

Fig. 7.1

(c) Using Fig. 7.1, calculate the mean percentage decrease per year in the number of MDR-TB cases between 2000 and 2010.

Show your working.

Answer = ……………………. % [2]

For Examiner's Use

(d) Scientists have concluded that the overuse of antibiotics will lead to an increase in MDR-TB cases.

Evaluate the validity of this conclusion using the data in Fig. 7.1.

[4]

[Total: 12]

8 Fig. 8.1 shows the position of two genes on three chromosomes II, III and IV of the fruit fly, *Drosophila melanogaster.*

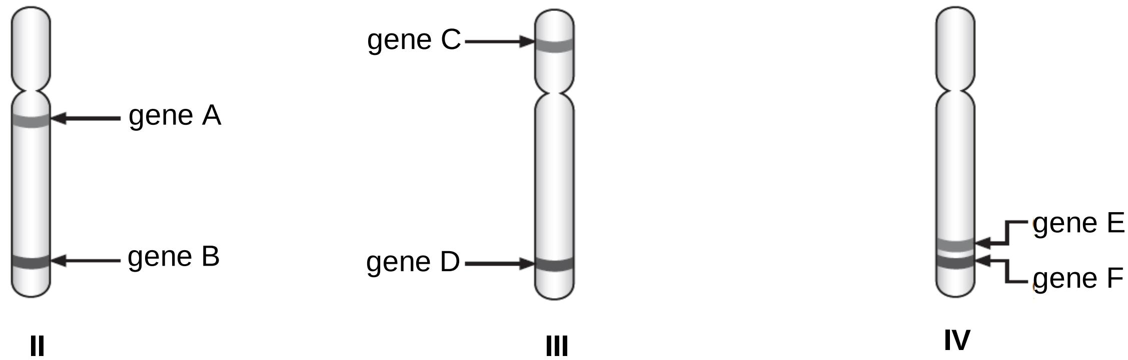

Fig. 8.1

(a) Explain which chromosome is most likely to show complete linkage between the two genes labelled in each chromosome.

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A group of geneticists examining inheritance in *D. melanogaster* found that gene A and gene B on chromosome II codes for body colour and wing type respectively (Fig. 8.2). The two genes are 28 cM (centimorgan) apart. When they crossed a pure-bred brown-bodied, curved-winged fly with a pure-bred black-bodied, normal-winged fly, all the F_1 offspring produced were observed to have brown body and curved wings. They then proceeded to test cross an F_1 offspring with a fly homozygous recessive for both traits.

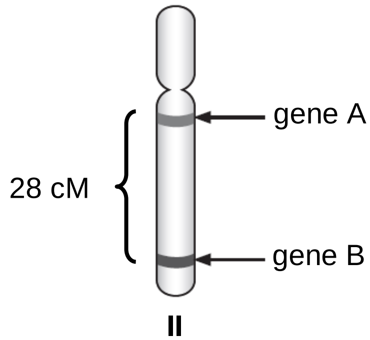

Fig. 8.2

(b) A recombinant frequency of 1% is defined as 1 cM distance between two genes on a chromosome. Recombinant frequency can be calculated using the equation:

$$\textit{Recombinant frequency} = \frac{\textit{Number of recombinant offspring}}{\textit{Number of total offspring}} \times 100\%$$

Table 8.1 shows the phenotypes observed in a total of 50 progeny from the test cross with the F_1 offspring.

Table 8.1

phenotype	number of offspring
brown body, curved wings	
brown body, normal wings	
black body, curved wings	
black body, normal wings	
total	50

Using Fig. 8.2 and the equation of recombination frequency, calculate the expected number of progeny with each phenotype and complete Table 8.1. [1]

For Examiner's Use

(c) Using the letters A/a and B/b to represent the alleles of the two genes, illustrate the test-cross with the use of a genetic diagram.

[4]

(d) Suggest how the allele for gene A gives rise to a brown body in *D. melanogaster*.

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[Total: 10]

For Examiner's Use

9 E-cigarette smoke (ECS) delivers nicotine as an aerosol without tobacco or the burning process. It does not contain carcinogenic by-products due to incomplete combustion and is thought to be less hazardous than combustible cigarettes.

To understand the carcinogenicity of ECS, a group of scientists investigated the effects of nicotine present in ECS on human lung epithelial cells.

Fig. 9.1a shows the relative DNA repair activity in varying concentrations of nicotine.

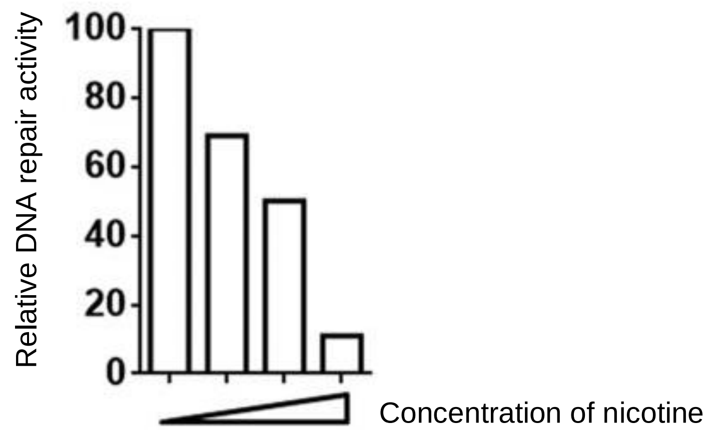

Fig. 9.1a

Fig. 9.1b is the ethidium bromide-stained gel electrophoregram showing the amount of DNA coding for the XPC protein produced in varying concentrations of nicotine. This is indicative of the abundance of XPC, a crucial protein involved in DNA repair.

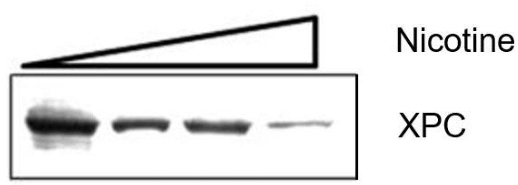

Fig. 9.1b

Colony formation on soft agar can be used to assess anchorage-independent growth. Fig. 9.2 shows anchorage-independent growth of lung cells on soft agar in the absence and presence of nicotine.

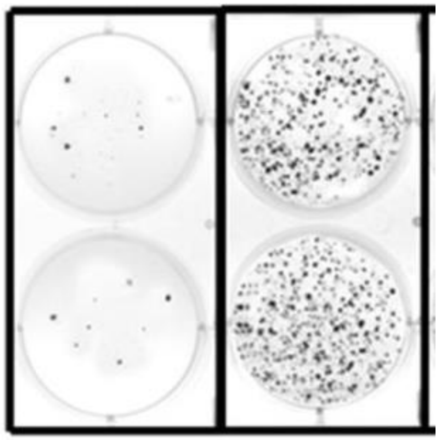

Fig. 9.2

(a)(i) Define the term cancer.

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(ii) With reference to Fig. 9.1 and 9.2, explain how ECS may promote lung cancer development.

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Nicotine is also known to increase the release of norepinephrine and epinephrine from nerve terminals. The binding of these chemicals to β-adrenergic receptors stimulates processes related to tumour progression as shown in Fig. 9.3.

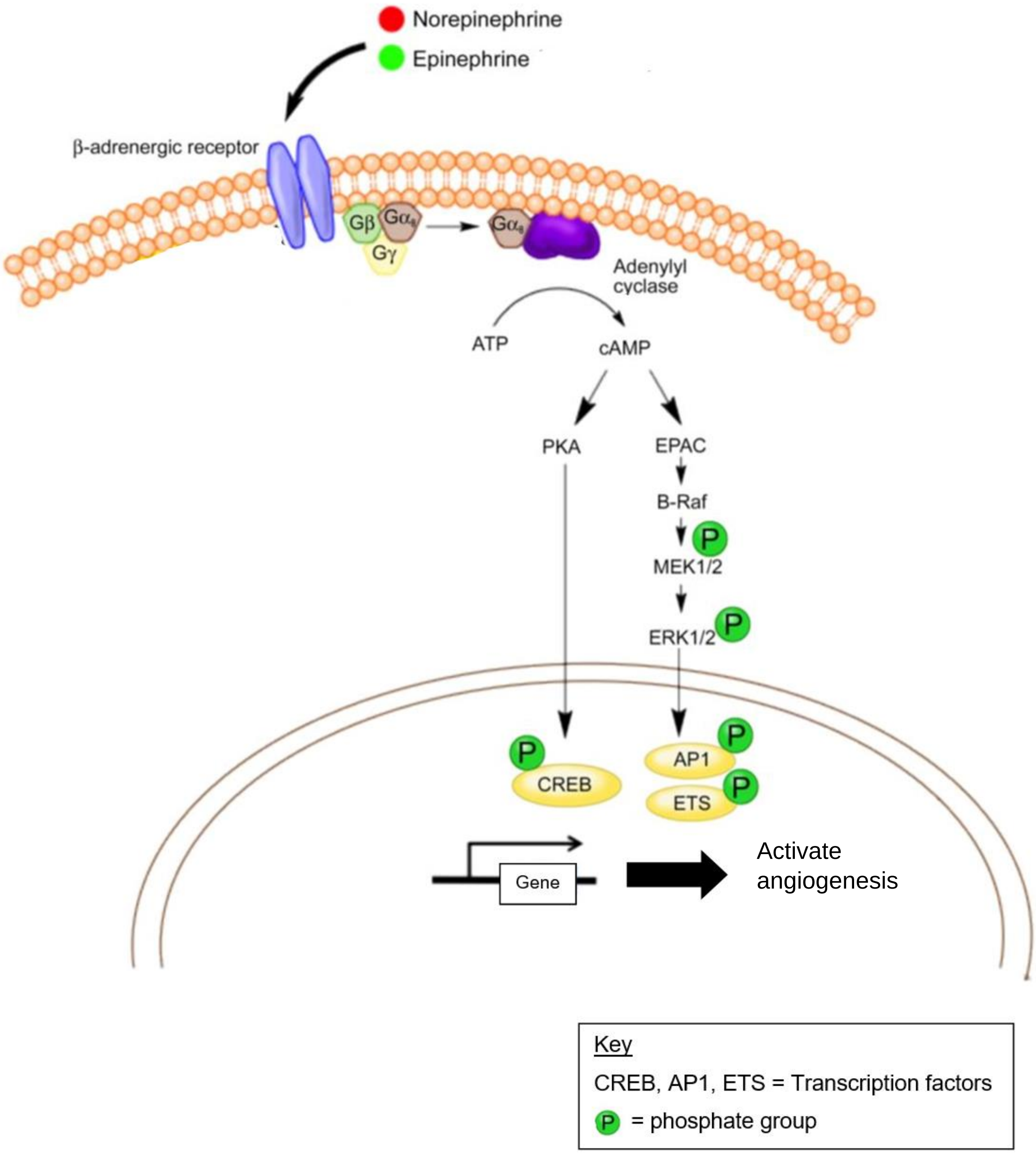

Fig. 9.3

(b) Proteins known as β-blockers are used specifically to bind and inhibit β-adrenergic receptors.

With reference to Fig. 9.3, explain how β-blockers can be used to treat cancer.

[4]

[Total: 10]

10 Climate change is one of the biggest concerns in today's world. Increased temperatures and extreme weather events has had severe effects on plants and animals globally.

(a) Explain how human activities have led to increased temperatures.

[2]

Scientists are starting to see changes in plant morphology due to climate change. A recent study documented how more flowers have increased UV-absorbing pigmentation as protection against the sun's radiation. One example of such a flower is the alpine cinquefoil.

(b) Explain how high levels of UV-absorbing pigmentation in flowers became common in a population of alpine cinquefoil.

[5]

While having more UV-absorbing pigmentation helps protect the pollen of the flowers from damage, scientists are worried about the negative downstream effects.

Hoverflies, the flower's natural pollinators, navigate their way to pollen by sensing UV light. More UV-absorbing pigments changes the way hoverflies perceive flowers. Fig 10.1a and 10.1b show samples of alpine cinquefoil flowers collected 60 years ago and recently under UV light respectively. Dark petal areas possess UV-absorbing compounds whereas lighter areas are UV reflective.

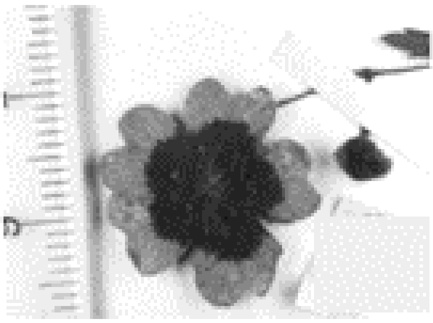

Fig 10.1a

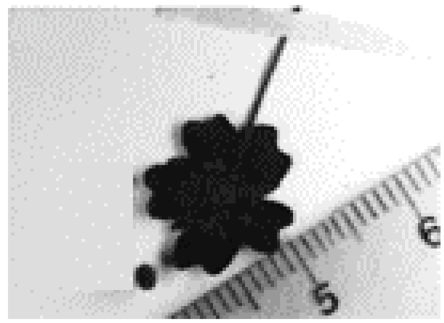

Fig 10.1b

(c) With reference to Fig 10.1, suggest how the change in pigmentation might affect the plant population.

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.... [2]

[Total: 7]

For Examiner's Use

11 The administering of COVID-19 vaccinations requires two shots – a first dose and a booster dose. Fig. 11.1 shows the share of people partly and fully vaccinated against COVID-19 in some countries over a period of five months until May 2021.

By May 2021, the daily new confirmed COVID-19 cases is 30 per million people in United Kingdom and 230 per million people in India.

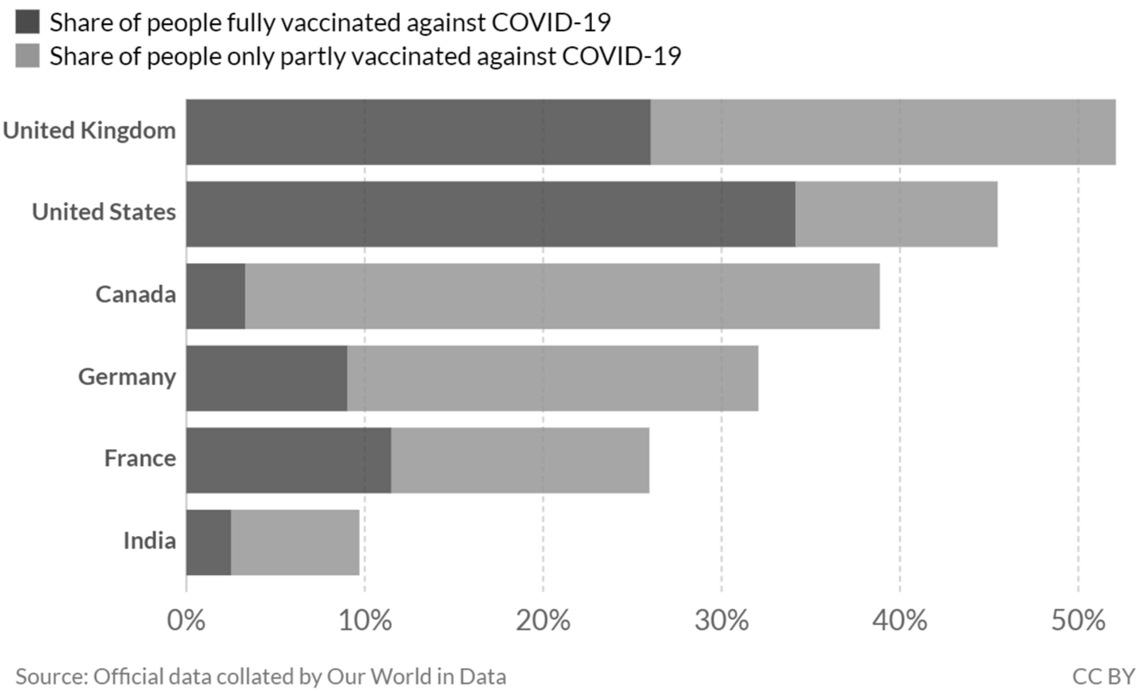

Fig. 11.1

(a) With reference to Fig. 11.1, explain why infection rate in UK is low while it is high in India.

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.... [3]

For Examiner's Use

(b) State two significant difference between the primary and secondary immune responses after the COVID-19 vaccinations.

[2]

[Total: 5]

END OF PAPER